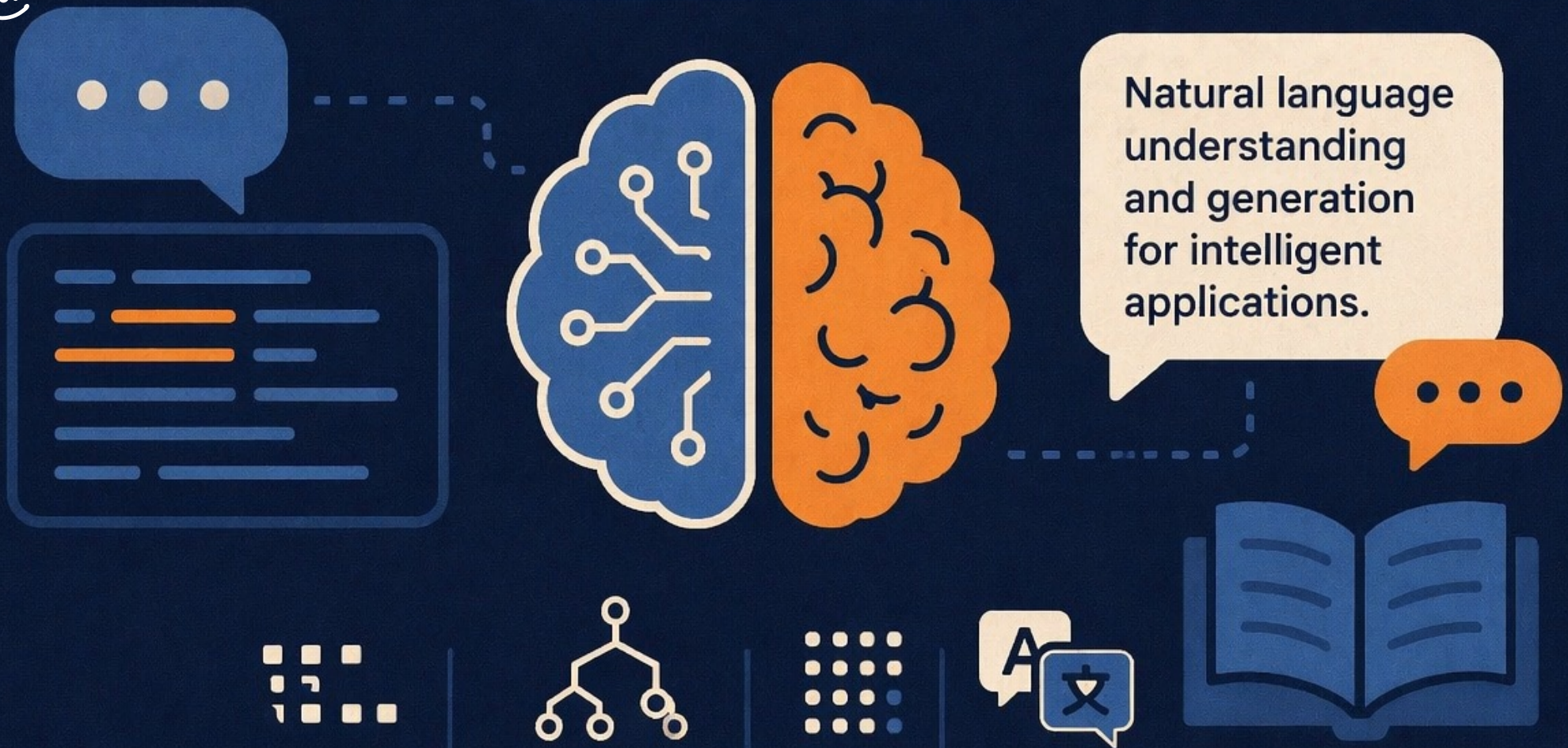
NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



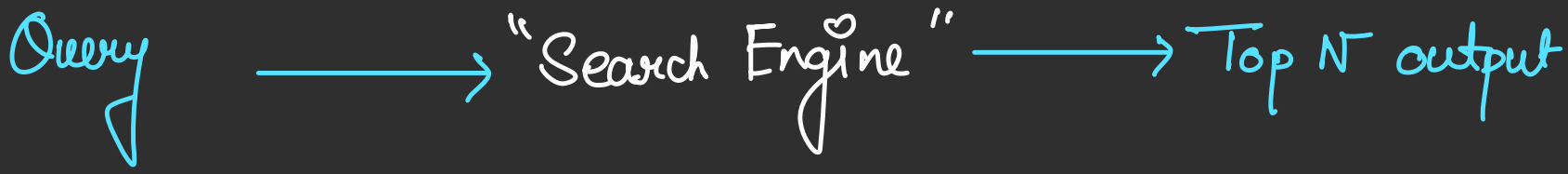
Lecture: Word Embeddings

#Agenda

1) Business Case Study
2) Contextual Word Embeddings
3) SVD Based methods
4) Word 2 Vec

problem Statements

"Over 50,000 publications & research paper on C19 till date"

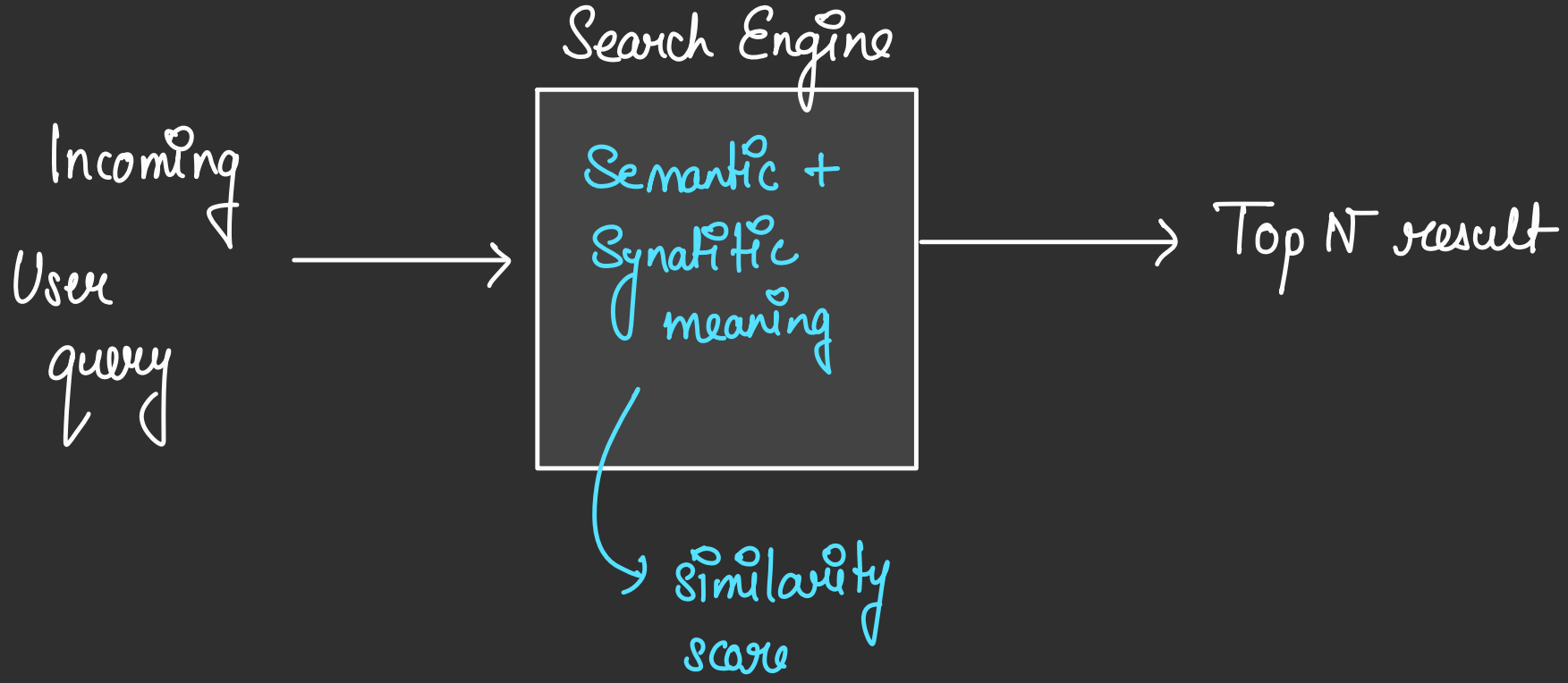


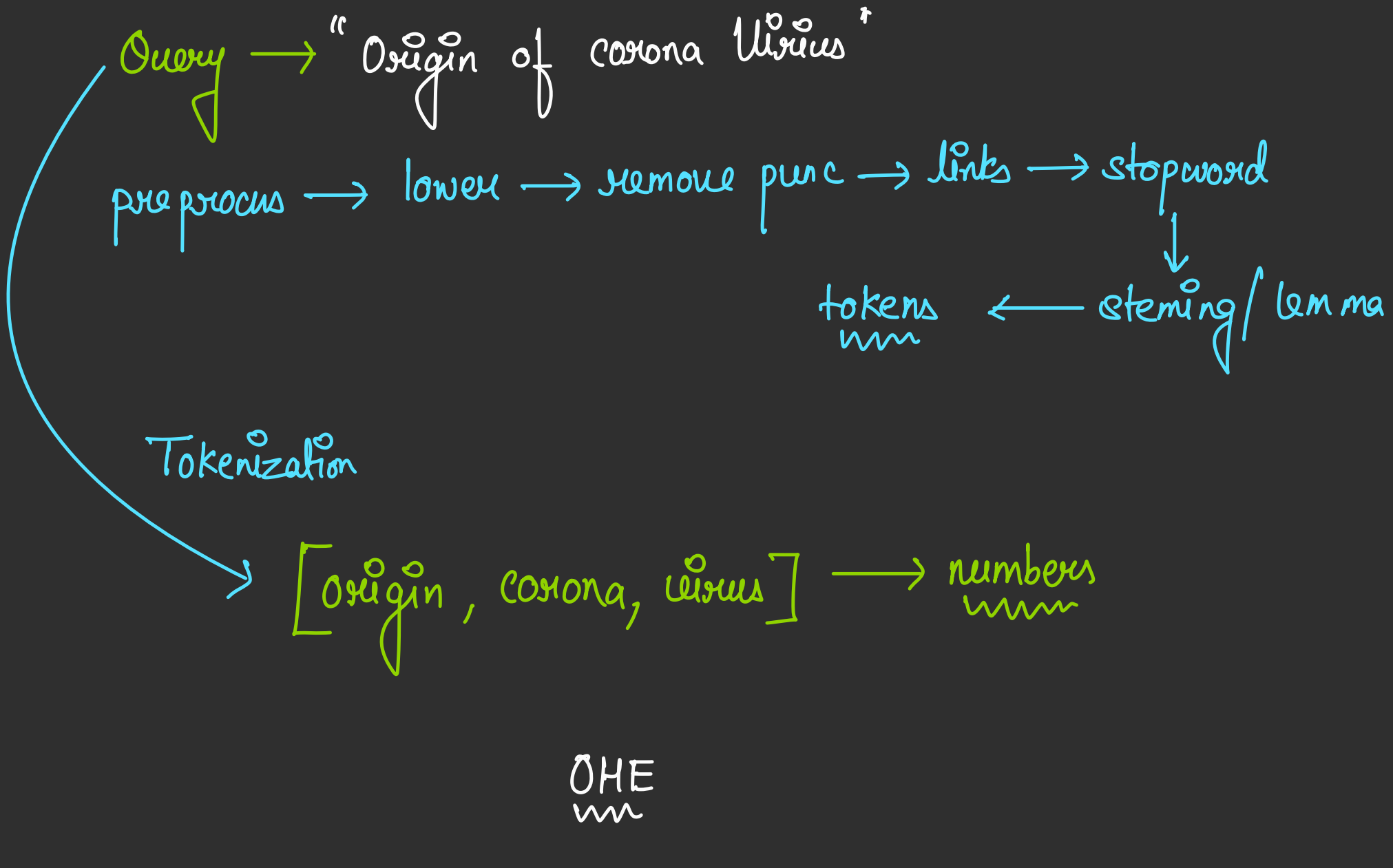
Challenges

1) Key word search (Exact matching)

Covid 19 + heath + origin

Architecture




Query → "Origin of corona Virus"
preprocess → lower → remove punc → links → stopword
tokens ← steming / lemma
Tokenization
[origin, corona, virus] → numbers
OHE

OHE

document

	Origin	...	Corona	....	Virus
Query	1				
Origin					
Corona					
Virus					

↳ Huge sparse matrix

No symantic relationship

SVD (Singular Value Decomposition)

$$M = U \Sigma V^T$$

$U \rightarrow$ captures the relationship b/w rows

$\Sigma \rightarrow$ singular values

$V \rightarrow$ captures the relationship b/w colours

Co-occurance Matrix

	cat	dog	milk	bark
cat	0	5	3	1
dog	5	0	1	6
milk	3	1	0	0
bark	1	6	0	0

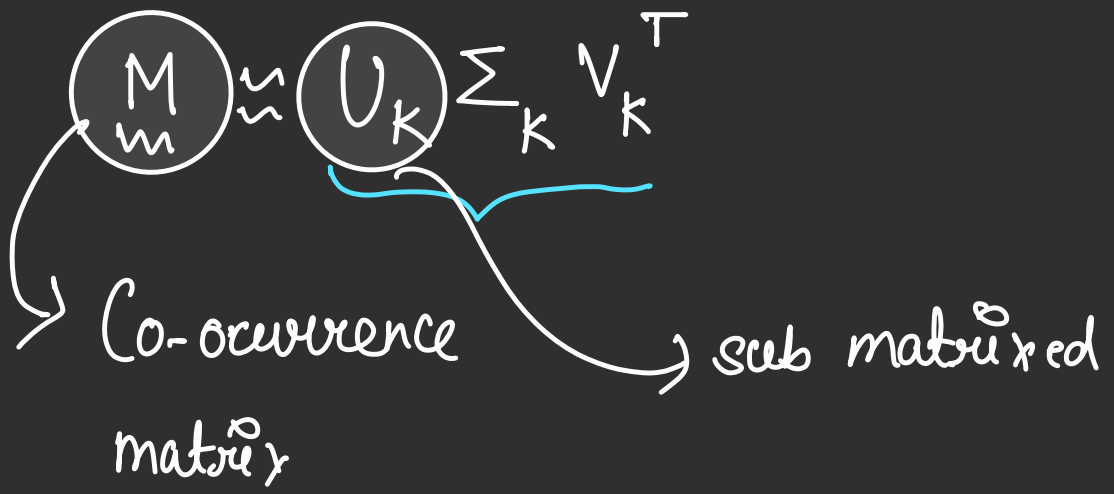
'How often words appear near each other'

Raw co-occurrence matrix

1) Huge
2) Sparse
3) Noisy

SVD helps to reduce dimentionality

We only keep top k values



$doc_1 \rightarrow$ [cat dog milk] (Word token)

$doc_2 \rightarrow$ [cat . book milk]

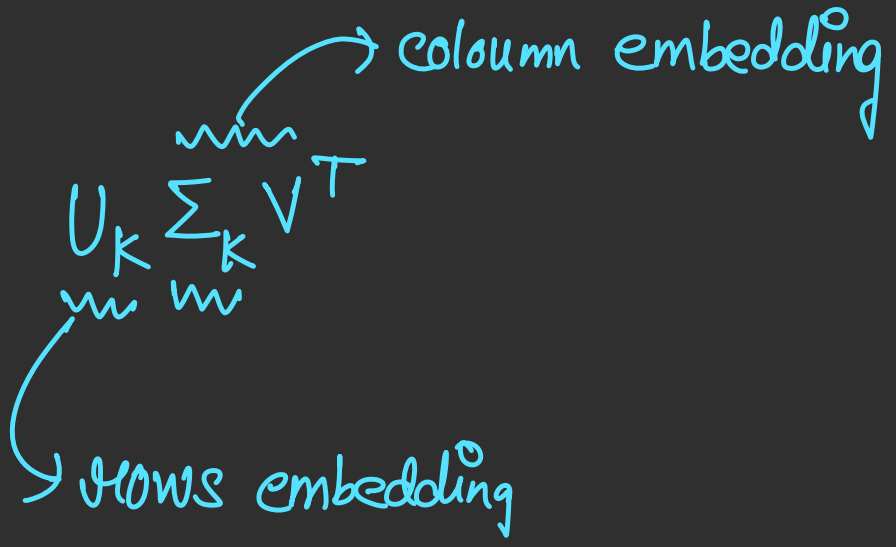
Steps for SVD

1) Compute SVD co-occurence matrix

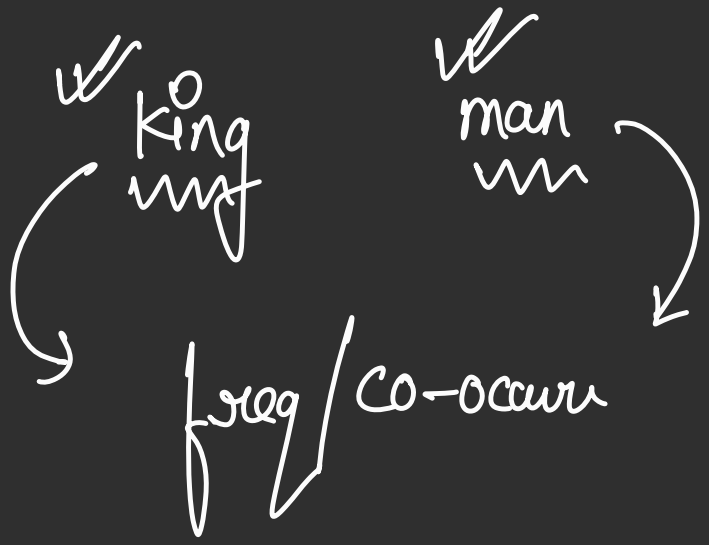
2) keep top k components

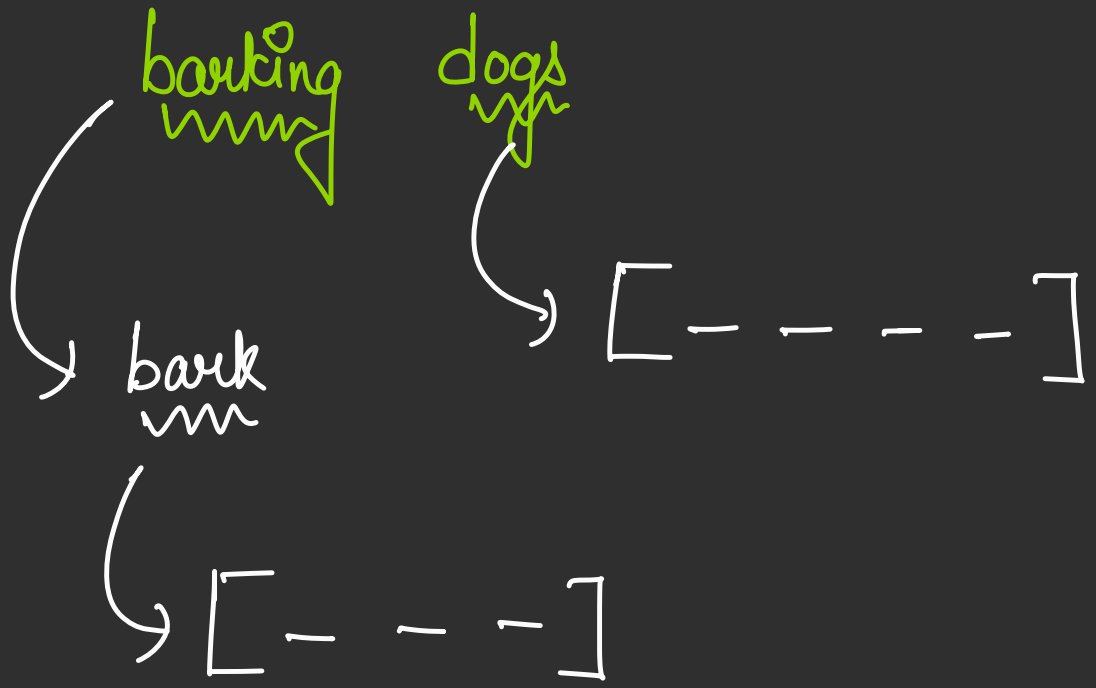
3) Use rows of $U_k \Sigma_k$ as embeddings

$k \longrightarrow$ parameters



SVD




barking
dogs
bark
[- - - -]
[- - -]


Word2Vec

The Core Idea

Method to turn words into dense vectors (Embeddings) such that

"Words with similar meanings stay close by"

Example:

king − man + woman ≈ queen

#Why word2Vec is a Bigdeal?

Before Word2Vec

1) Word → OHE
2) No notion of similarity
3) High dimentional vocal size

Word2Vec

1) Words → dense vectors (100-300)
2) Captures:
   - semantic similarity (king v queen)
   - syntatic similarity (run v running)

Types of Word2Vec

1) CBOW (Continuous Bag of Words)

predicts the target word from surrounding context

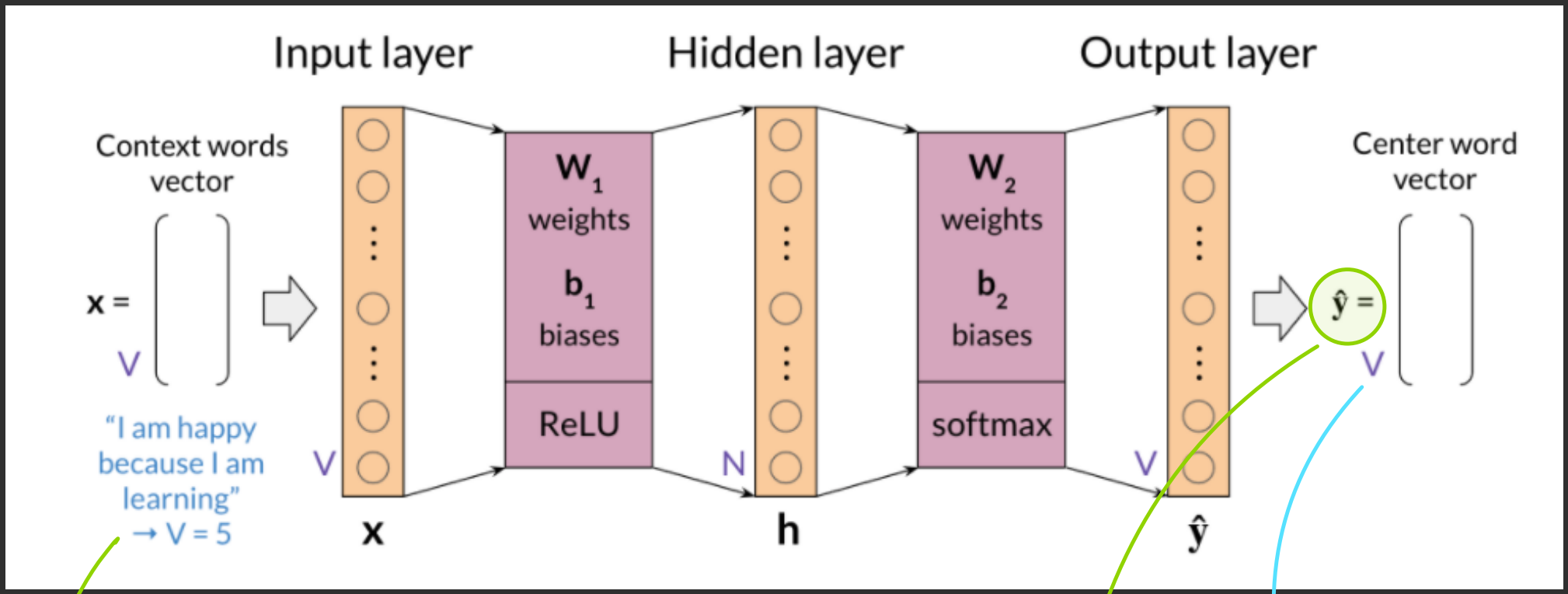
Ex: The cat ___ on the mat

Target: sat

Behaviour

→ Faster

→ Works well for frequent words



Shallow NN

OHE

I = [1 0 0 0 1]

happy = [0 1 0 1 1]

softmax

prob of all words from the vocabulary

(Multi-class)

loss fun^n ⇒ CCE

sat → y true

The cat ___ on the mat

Input: The, cat, on, the, mat (OHE)

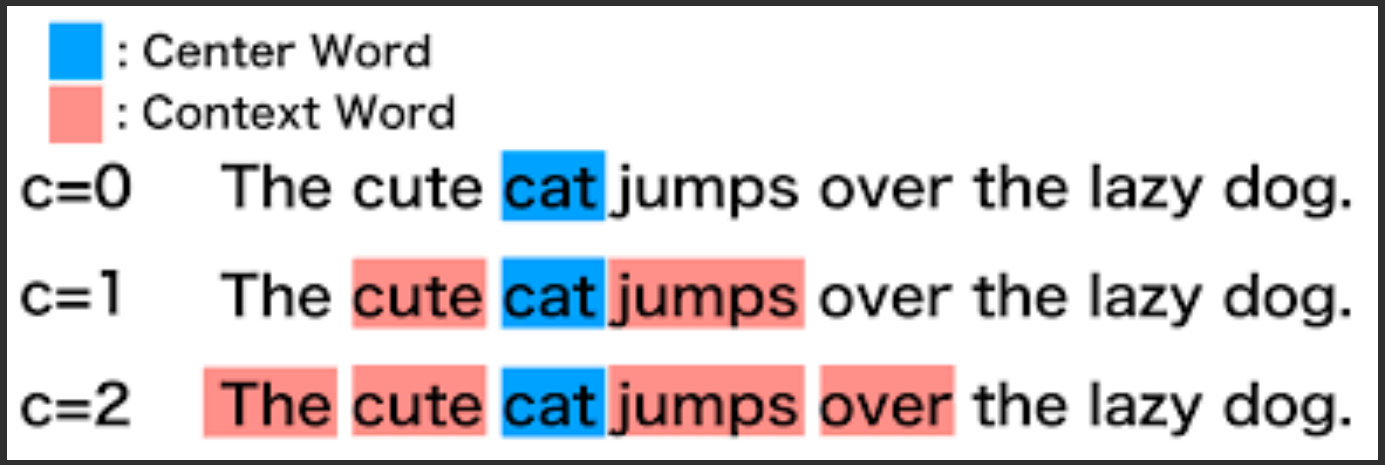
Output: sat: 0.89

cat: 0.01

dog: 0.02

on: 0.86

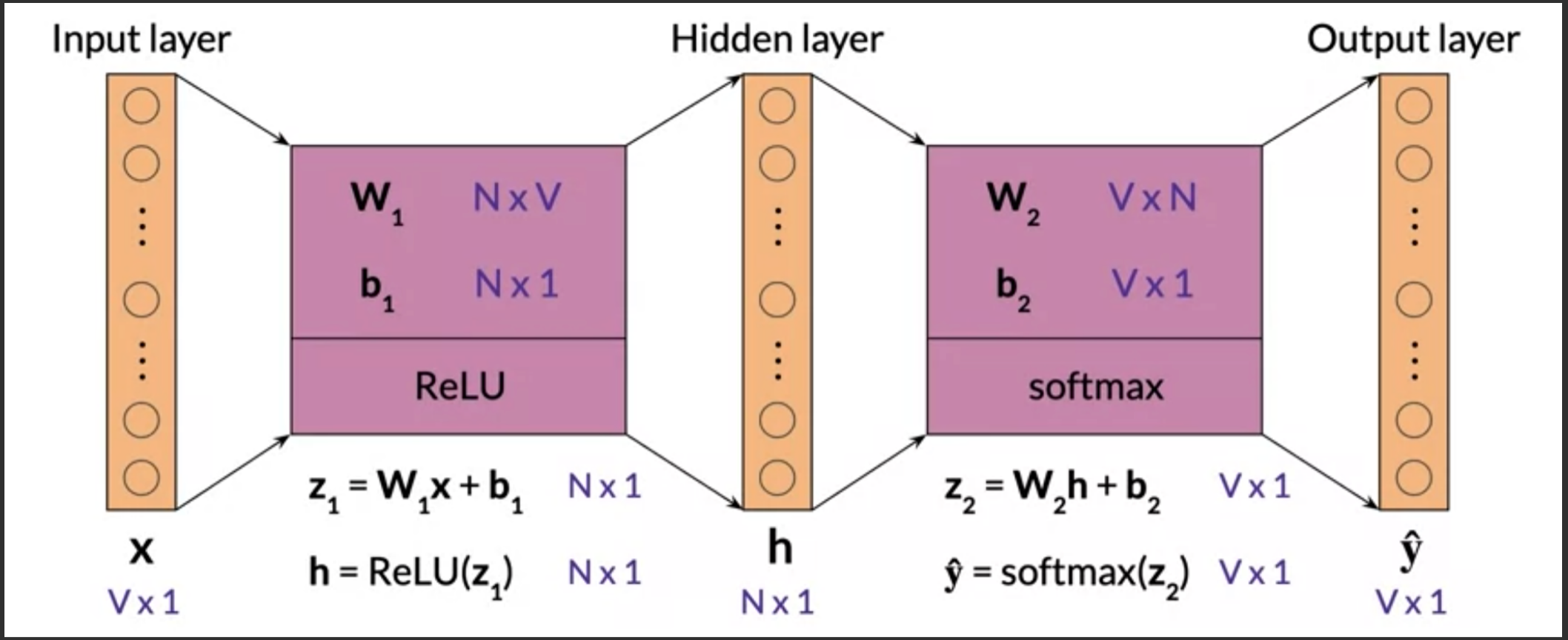
How do we generate training data for CBoW?



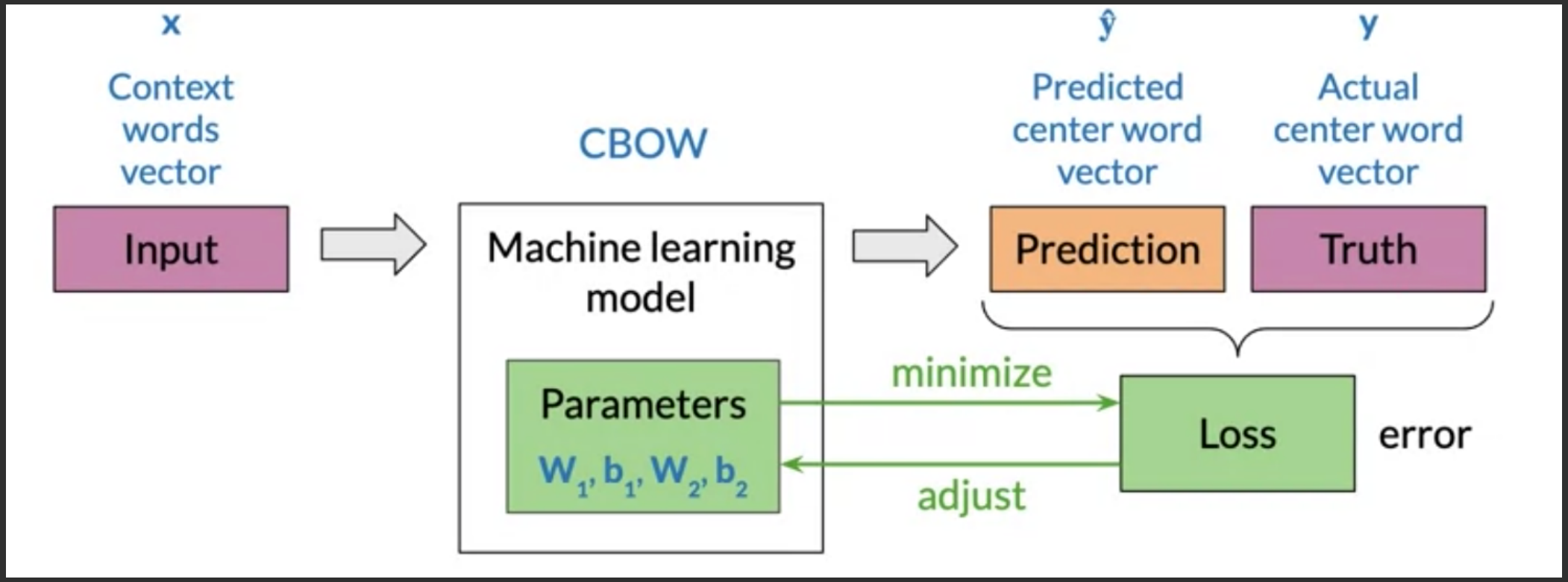
Training data is generated by iterating through the corpus using a sliding window action to define the target word and its context words.
The window size (size of the sliding window) determines the number of before and words to consider when predicting the target word.

Working of CBoW

CBoW models the problem as a multi-class classification, wherein given the context words, the model outputs the target word and in doing so learns the representation for the words.



How does the model learn the right word representations?



2) Skip-gram

predicts context words from target word.

Ex,

Target : sat

predict : cat, on, the, mat

Idea

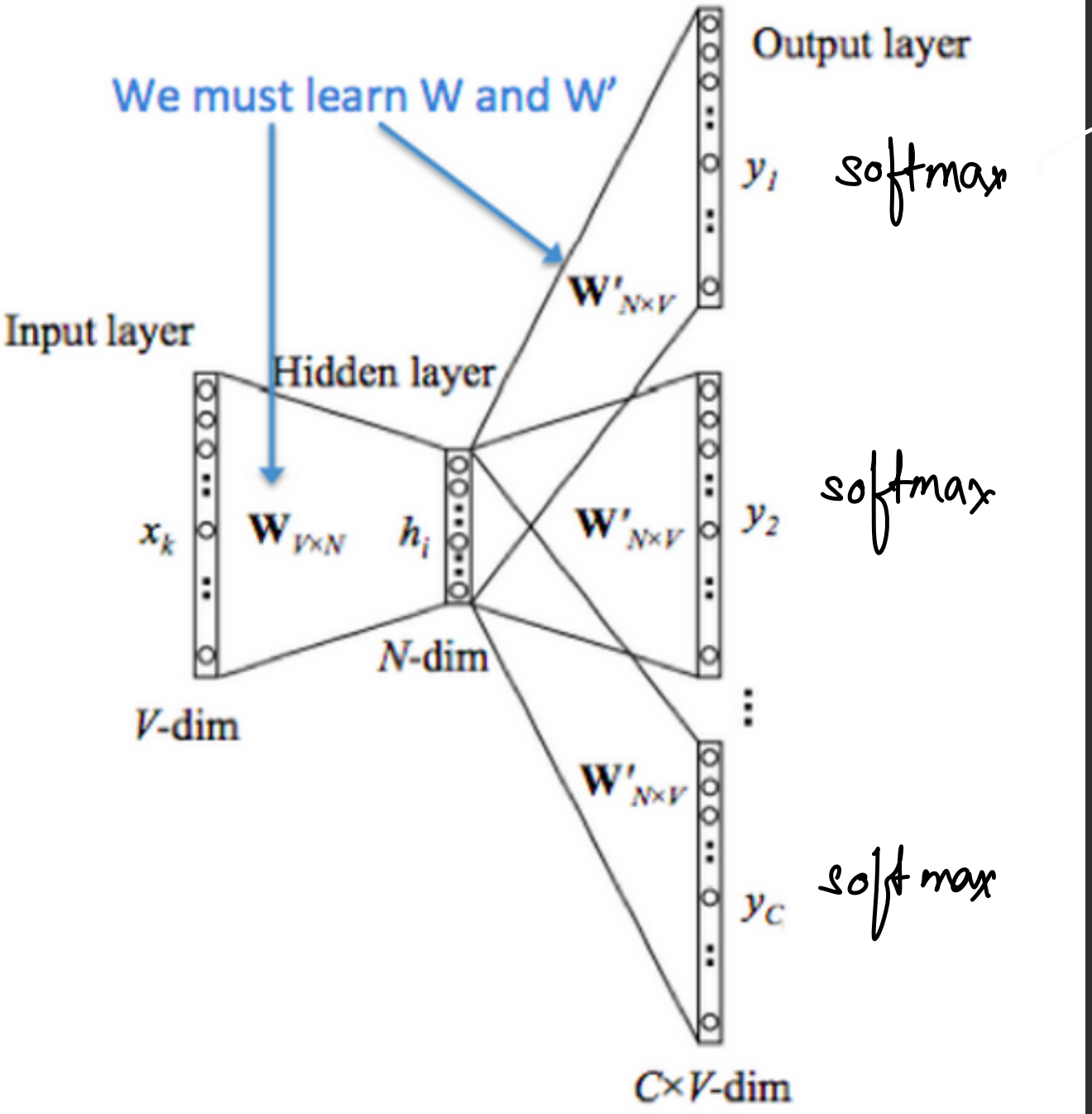
Input: One word

Output: surrounding word

Behaviour

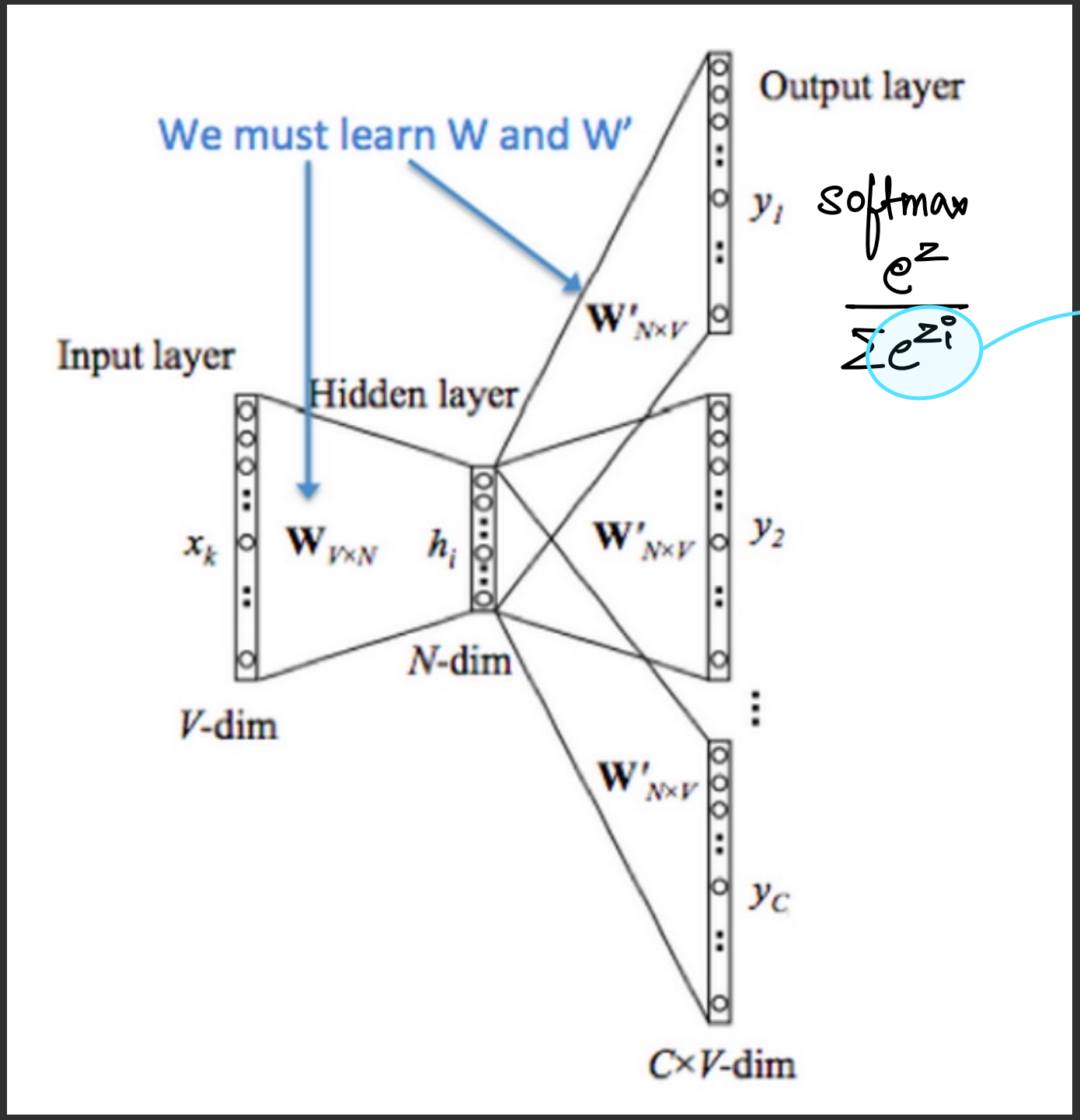
→ Better for rare words

→ Slower than CBOW



→ sum of all prob

softmax



$V = 100{,}000$

↳ Vocab

softmax

$$\frac{e^{z}}{\sum e^{z_i}}$$

→ 100,000 V

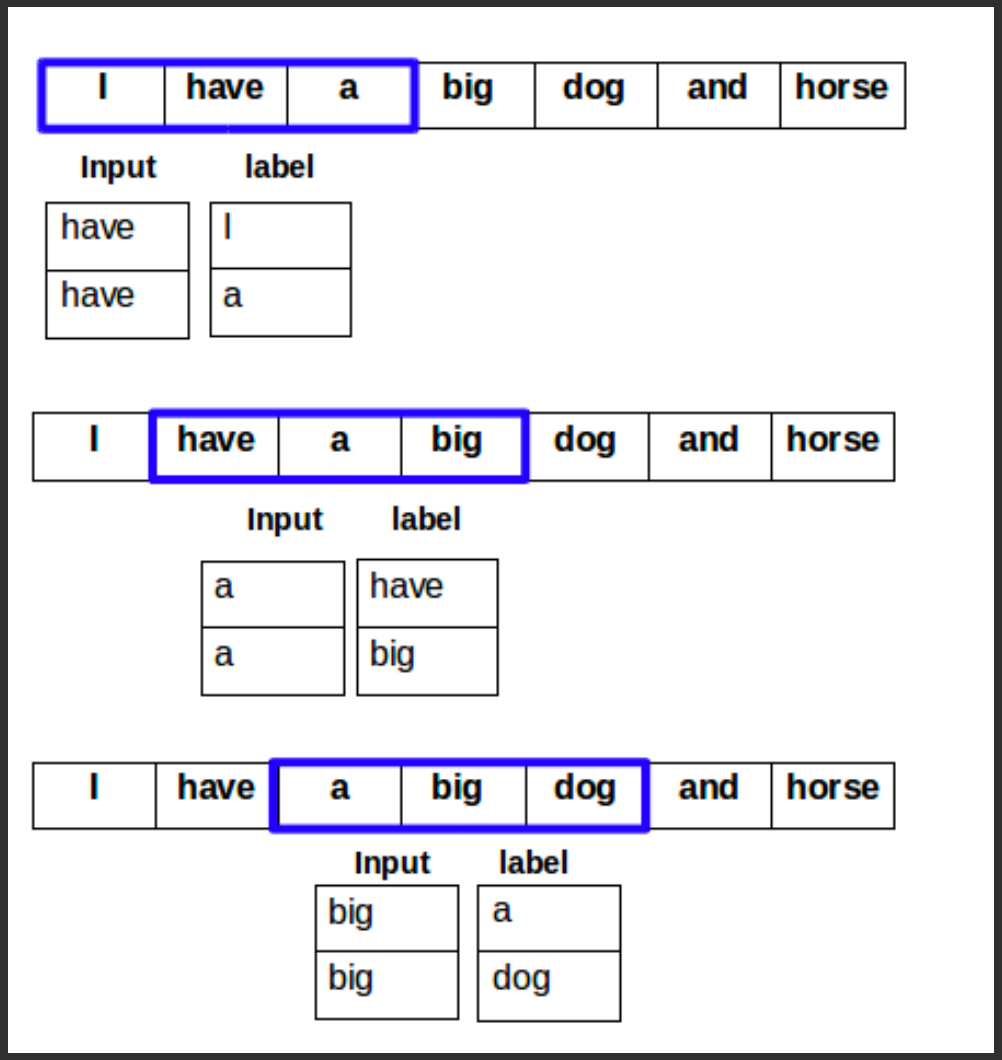
Context = 50

↳

50 x 100,000

How do we generate training examples for Skip-Gram?

Similar to CBoW, we generate training examples by iterating over the sentence. Based on window size, single word is chosen as input and its surrounding words would be the target.



Let say, C = 50

softmax :- 50

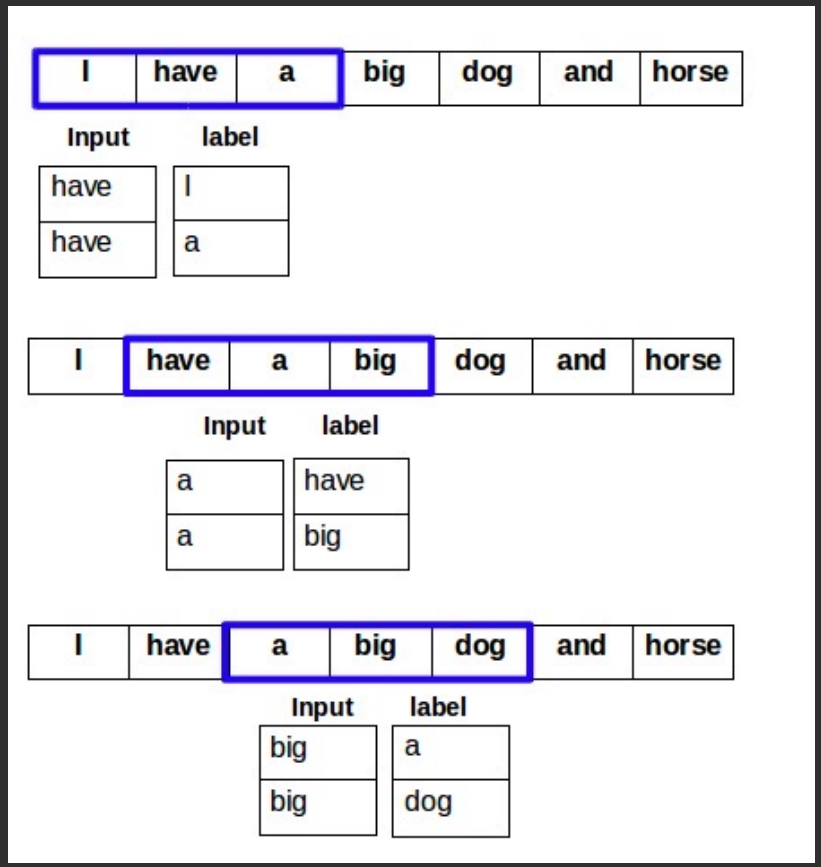
Negative Sampling

Softmax

↳ problem

computationally Expensive

{ 1 positive sample
k negative samples



{ have → Input
1 → output
a → output

↳ (1, have)

↳ pos sample

(big, dog)
(and, horse)
(a, big)

{ k negative sample

Objective Fcn =

$$\log \sigma\left(v_{w_o}^{T} \cdot v_{w_i}\right) + \sum_{K} \log \sigma\left(-v_{w_K}^{T} \cdot v_{w_i}\right)$$

pos sample

K Neg sample

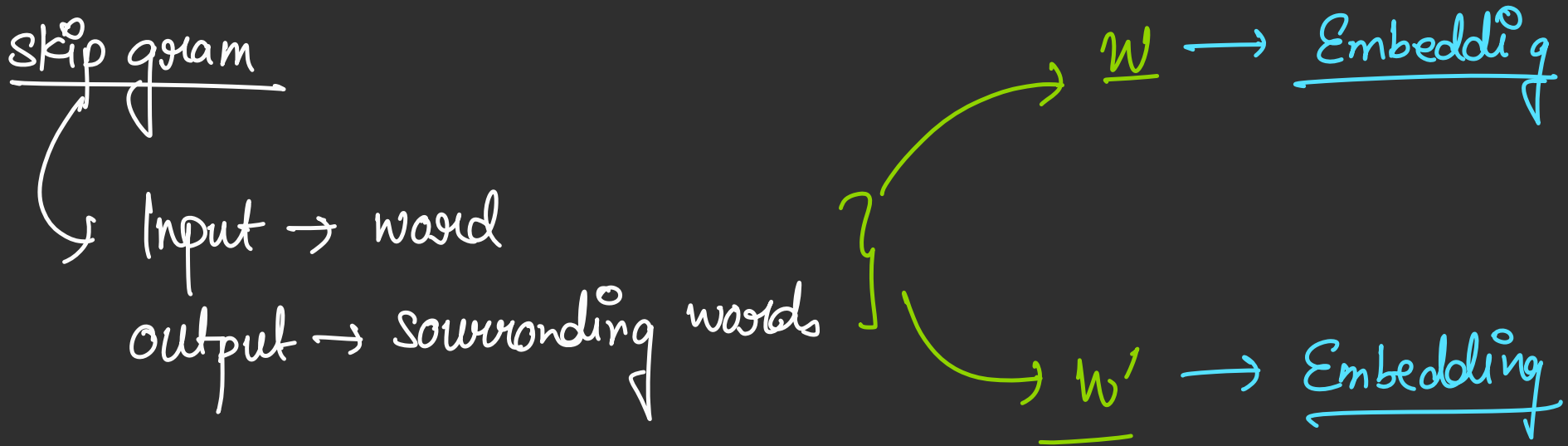
Intuition

→ Push correct word closer

Push random word away

Binary class

→ sigmoid


skip gram
Input → word
output → sourrounding words
W → Embeddi g
W' → Embedding

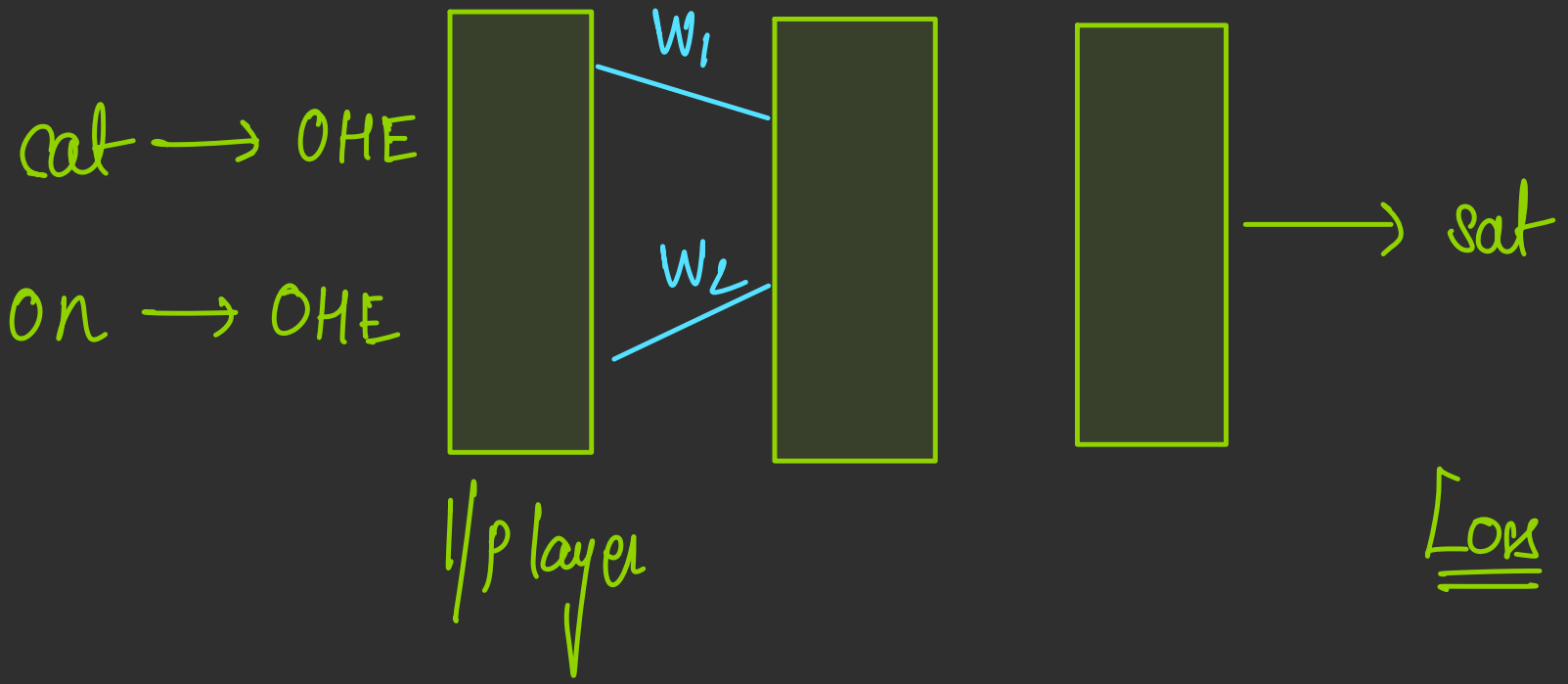
Doubts

cat ___ on

y true: sat
y pred: mat

cat ___ on the mat

↳ sat / mat / dog /



sigmoid

→ Binary classification

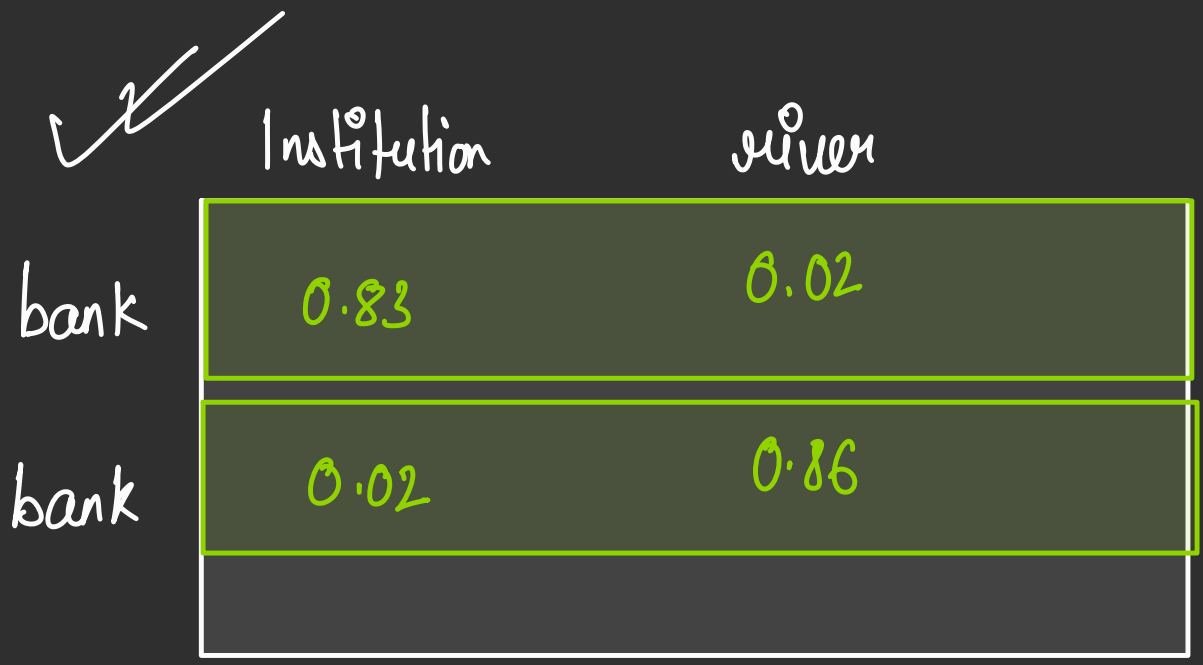
$$a(x) = \frac{1}{1 + e^{-x}}$$

softmax

→ Multiclass classification

$$\frac{e^{z}}{\sum_{i=1}^{n} e^{z_i}}$$

→ Total probability


Institution
river
bank
0.83
0.02
bank
0.02
0.86

OHE